NAGARESCUE

FIGMALINK: NAGARESUE

HCI TEMPLATE

MEMBERS:

Francisco, Bryann Joshua
Abada, Rachelle Ann
Diego, Gabriel David
Bo, Daryl Adrian

Barangay Rescue and Early Warning System (B.R.E.W.S)

Bo, Daryl Adrian

USER PERSONA:



Lloyd Fabricate

Rescuer

Resilient | Caring | Family-oriented

Age: 43

Location: Barangay Igualdad, Zone 5, Naga City, Camarines Sur, Philippines

Biography

Lloyd is a dedicated field rescuer in Naga City with nearly 8 years of frontline experience in disaster response. He is trained in first aid, water search and rescue, and navigation, making him one of the most relied-upon rescuers during typhoons and floods. Lloyd has been deployed in both urban centers and remote barangays, where access is often difficult due to rising waters, blocked roads, or landslides.

During Typhoon Kristine in October 2024, Lloyd was among the first responders deployed in severely flooded areas of Barangay Igualdad and nearby communities. He faced the challenge of wading through chest-deep waters, rescuing families trapped on rooftops, and navigating without clear route data due to poor signal and overwhelmed communication lines. The lack of equipment and the overwhelming number of SOS calls left him and his team exhausted, but they continued working non-stop, driven by the urgency to save as many lives as possible.

Challenges

- Difficulty accessing stranded areas due to high floodwaters, blocked routes, and damaged infrastructure.
- Lack of real-time route and hazard data, leading to delays and risks during operations.
- Shortage of proper rescue equipment (boats, life vests, ropes, radios) during major typhoons.
- Overwhelming number of requests for help during peak disasters, causing physical and mental fatigue.
- Uncoordinated or unclear instructions from command centers slow down field operations.

Technology Proficiency

- Proficient in using smartphones for communication (calls, SMS, Messenger, Viber).
- Familiar with navigation apps (Google Maps) but struggles when there is no internet or GPS signal.
- Basic use of rescue radios, but prefers mobile apps for faster updates.
- Comfortable using group chats for coordination, though overwhelmed when too many messages flood in.

Goal & Needs

- Have a reliable system with safe, updated routes during floods.
- View and respond to rescue requests through a centralized platform.
- Minimize delays in dispatching by connecting victims, barangays, and CDRRMO.
- Maintain clear, fast communication with CDRRMO and fellow rescuers.
- Reduce physical strain and risks through better coordination and resource allocation.

Personality



JOURNEY MAP:



Lloyd Fabricate

Age: 32
Occupation: Rescuer
Location: Barangay Cararayan, Zone 3, Naga City

PERSONALITY

- •Introverted: 8/10
- •Optimistic: 8/10
- •Calm: 8/10
- •Independent: 8/10

GOALS & NEEDS

- Have a reliable system with safe, updated routes during floods.
- View and respond to rescue requests through a centralized platform.
- Minimize delays in dispatching by connecting victims, barangays, and CDRRMO.
- Maintain clear, fast communication with CDRRMO and fellow rescuers.
- Reduce physical strain and risks through better coordination and resource allocation.

CHALLENGES/ PAIN POINTS

- Difficulty accessing stranded areas due to high floodwaters, blocked routes, and damaged infrastructure.
- Lack of real-time route and hazard data, leading to delays and risks during operations.
- Shortage of proper rescue equipment (boats, life vests, ropes, radios) during major typhoons.
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- Uncoordinated or unclear instructions from command centers slow down field operations.

TECHNOLOGY PROFICIENCY

- Proficient in using smartphones for communication (calls, SMS, Messenger, Viber).
- Familiar with navigation apps (Google Maps) but struggles when there is no internet or GPS signal.
- Basic use of rescue radios, but prefers mobile apps for faster updates.
- Comfortable using group chats for coordination, though overwhelmed when too many messages flood in.

Rescuer responding for SOS Alert

PHASES	Preparation and Alert	Receiving and Assessing Distress SOS	Deployment and Navigation	Rescue Execution	Standby for Next Rescue
ACTIONS	Lloyd, always rely on text messages, calls, or social media. However, when the internet and network are interrupted, citizens have no way to call for rescue.	When they are dispatched, they don't know how many people they need to rescue. Additionally, if someone requires specific medication, they may not be prepared to provide it.	They often get lost because they don't have the specific location, or sometimes they circle the area repeatedly to find the citizens who need their help.	Some evacuation center are over populated, and sometimes the citizen they rescued is not getting a comfort they want because of over crowded.	They dont have a break, even though they are exhausted, hungry, or even when they also need a aid.
EMOTIONS	Optimistic but focused, feeling a sense of duty	Empathetic and reliable, relieved by clear info	Calm and independent, motivated by helping others	Optimistic and community-oriented, feeling accomplished but exhausted	Calm and introspective, hopeful yet ready for action
SOLUTION IDEAS	SOS app sends push notifications with verified alerts and pre-loaded maps, allowing one-tap readiness confirmation	App centralizes SOS with auto-verified details, offering one-tap status updates and details of user to view what aid and medication we need to consider	App integrates real-time location tracing and flood-safe routes, displaying household details upfront for path optimization	App traces locations in real-time, auto-alerting teams for support, with household details for quick prep and post-rescue logging.	App sends updates to the CDRRMO, provides a rest schedule feature.

STORY BOARD:

Their Office is Overwhelmed with the Distress call from every barangay in their city



They are going insane because they are only 4 On Duty Officer



Their office has a dashboard that monitor every barangay status. And can deploy a rescuer with details



He is being dispatch to rescue a victims



Surprised to dispatch



Now, he knows every details of the victim. and now he can prepare



Finding the path to the victim and possible route



He is confused and going insane because he is not familiar with the place



No More checking of maps, The geolocation can guide him to the victims



Choosing Equipment and Tools to Use



Dont know what tools and equipment to bring because he dont know what are the status of the place and victims



He brings the victim to the Evacuation Center and provide their needs



After the rescue, he is assign again to a new rescue



Tired and Hungry he just came back from his previous mission



Now, he can update his status if he need a break, and ready for a new mission



USER & SYSTEM REQUIREMENTS SPECIFICATION

USER 1: BDRRMC

Barangay officials need a clear, distributed mechanism for managing and responding to emergency alerts that minimizes the load on the central CDRRMO office.

USER REQUIREMENTS

The system must provide barangay officials with a dashboard and tools to manage SOS alerts, deploy responders, and escalate cases to the CDRRMO when necessary, ensuring efficient local response and reduced dependency on central command.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
  - **FR1.1:** The system must display a dashboard listing all SOS alerts within the barangay, including user information, location, emergency type, and urgency.
  - **FR1.2:** The system must allow officials to assign available rescuers to specific SOS alerts and track their status.
  - **FR1.3:** The system must enable filtering and sorting of alerts by urgency, time, or status for efficient prioritization.
  - **FR1.4:** The system must provide an escalation option to forward complex cases to the CDRRMO, including all relevant alert data.
  - **FR1.5:** The system must generate reports summarizing resolved alerts for barangay-level record-keeping.
- **Non-Functional Requirements:**
  - **NF1.1:** The dashboard must update automatically in real-time without requiring manual refresh.
  - **NF1.2:** The system must remain responsive on both mobile and desktop devices.
  - **NF1.3:** Task assignments and escalations must occur within seconds to ensure rapid response.
  - **NF1.4:** Data transfers between the barangay and CDRRMO systems must be secure and seamless.

USER 2: CITIZEN

The resident user must be able to receive timely, location-specific alerts and send emergency (SOS) notifications when in need of assistance. This ensures they stay informed and can request help efficiently during emergencies.

HCI TEMPLATE

USER REQUIREMENTS

The system must provide a mobile application that delivers real-time, geographically tailored alerts to residents and allows them to send SOS alerts with their location and emergency details to barangay officials for immediate response.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
  - **FR2.1:** The system must display alerts containing details such as hazard type, severity, location, and recommended actions.
  - **FR2.2:** The system must allow residents to view active alerts and their current statuses.
  - **FR2.3:** The system must enable push notifications for immediate delivery of alerts to mobile devices.
  - **FR2.4:** The system must allow residents to view nearby evacuation centers.
  - **FR2.5:** The system must allow residents to send an SOS alert through a triple-tap gesture, automatically including their current or designated location.
  - **FR2.6:** The system must enable users to specify the type of emergency, urgency level, and number of affected individuals (e.g., household members).
  - **FR2.7:** The system must confirm to the user that the SOS alert has been successfully sent and is being processed.
- **Non-Functional Requirements:**
  - **NF2.1:** Alerts and SOS messages must be delivered quickly and reliably, even under low-network conditions.
  - **NF2.2:** The system must remain operational, stable, and secure at all times.
  - **NF2.3:** The interface must be simple, intuitive, and accessible to users with limited technical skills or during stressful conditions.
  - **NF2.4:** User data, including location and personal information, must be protected to ensure privacy and security.

USER 3: CDRRMO

It should minimize the load on the CDRRMO office so they’re not going to be overwhelmed. And receive a request from the BDRRMC to support them in terms of the BDRRMC getting overwhelmed or a lack of a Rescue team.

USER REQUIREMENTS

The dashboard should receive a notification from the BDRRMC to help them with their rescue operation. And the CDRRMC can deploy a rescuer from their S.A.R.U. or ComCen team.

HCI TEMPLATE

SYSTEM REQUIREMENTS

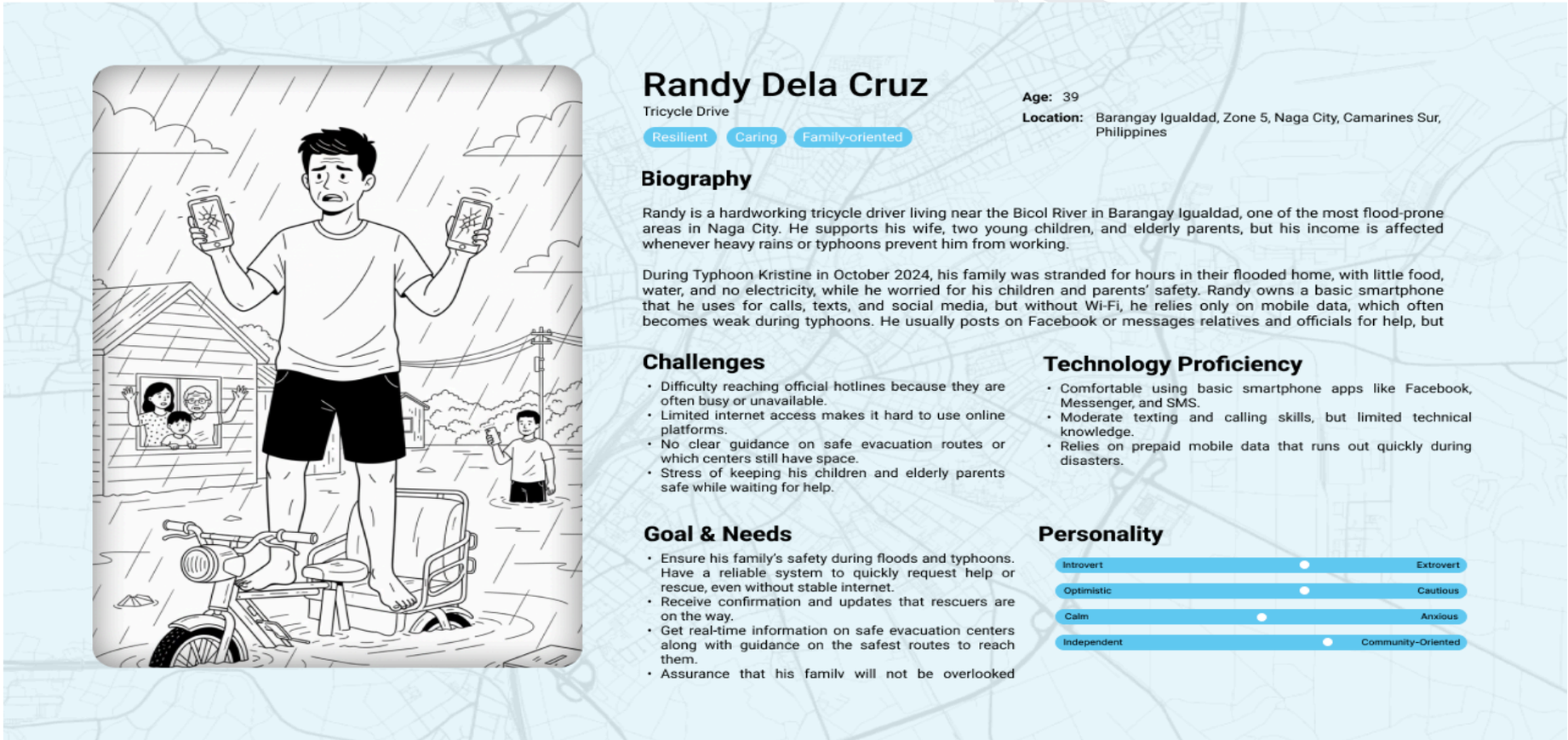
- **Functional Requirements:**
  - **FR3.1:** The system must display a dashboard listing all escalated SOS alerts from barangays, including user information, location, emergency type, and urgency.
  - **FR3.2:** The system must allow assignment of available rescuers to specific SOS alerts and track their operational status.
  - **FR3.3:** The system must enable filtering and sorting of alerts by urgency, time, or status for efficient prioritization.
  - **FR3.4:** The system must generate reports summarizing resolved and assisted alerts.
  -
- **Non-Functional Requirements:**
  - **NF3.1:** The dashboard must update automatically without requiring manual refresh.
  - **NF3.2:** The system must remain responsive across both mobile and desktop devices.
  - **NF3.3:** Task assignments and escalations must occur within seconds to ensure rapid response.
  - **NF3.4:** Data transfers between the barangay and CDRRMO systems must be secure and seamless.

HCI TEMPLATE

Residents Rescue Request Management System (RRRMS)

Rachelle Ann Abada

USER PERSONA:



JOURNEY MAP:



Randy Dela Cruz

Age: 38
Occupation: Tricycle Driver
Location: Barangay Igualdad, Zone 5, Naga City

PERSONALITY

- •Introverted: 4/10
- •Cautious: 8/10
- •Anxious: 5/10
- •Community-Oriented: 8/10

GOALS & NEEDS

- Ensure his family's safety during floods and typhoons.
- Have a reliable system to quickly request help or rescue, even without stable internet.
- Receive confirmation and updates that rescuers are on the way.
- Get real-time information on safe evacuation centers along with guidance on the safest routes to reach them.
- Assurance that his family will not be overlooked during rescue operation

CHALLENGES/ PAIN POINTS

- Difficulty reaching official hotlines because they are often busy or unavailable.
- Limited internet access makes it hard to use online platforms.
- No clear guidance on safe evacuation routes or which centers still have space.
- Stress of keeping his children and elderly parents safe while waiting for help.

TECHNOLOGY PROFICIENCY

He is comfortable using basic apps like Facebook, Messenger, and SMS, but with limited technical knowledge, relying on prepaid mobile data that often runs out during disasters.

Randy during before, during, and after Typhoon Kristine

PHASES	Before the Flood (Warning Stage)	During the Flood (Emergency Rising)	Rescue Coordination (Waiting for Help)	Rescue Arrives	At the Evacuation Center	Recovery & Returning Home
ACTIONS	• After hearing the news from his neighbors about the typhoon, he tries to prepare by moving some things to higher shelves.	• When water levels rise quickly, Randy tries calling hotlines but the lines are busy. • He posts on Facebook about their situation and messages someone from the barangay to ask for help.	• He keeps checking his phone but is unsure if rescuers see his post. • He tries calling again but still gets no answer. • He decides to move his whole family to the roof since their house has no second floor.	• Rescuers finally arrive after 6 hours. • He helps his family board the boat, and the evacuation takes longer because of blocked and flooded roads.	• He registers his family at the evacuation center. • He looks for food, water, and medical support. • He tries to contact relatives to let them know they are safe.	• They returns after the flood subsides, cleans the damaged house, checks their belongings, and restarts work as a tricycle driver. • He faces no financial aid, little recovery support, and an ongoing flood risk.
EMOTIONS	He feels worried about his family's safety.	He feels frustrated by the lack of response and panics as the floodwater rises since their house is low.	He is afraid that rescuers might not come; he feels a bit hopeless but still clings to survival.	He feels relieved but very tired.	He feels safe but still anxious about their home.	He is tired, overwhelmed, but determined to rebuild.
SOLUTION IDEAS	• Early warning alerts through a centralized mobile app that can still be accessible offline.	• One-tap SOS request system that works even with weak data or SMS. • Auto-send GPS location + number of people + urgency level to CDRRMO. • Automatic confirmation: "Rescue on the way."	• Real-time rescue tracking • Automatic SMS/app confirmation ("Rescue on the way"). • Priority alerts for families with children/elderly stuck on rooftops.	• Navigation system for rescuers with real-time safe route mapping. • Centralized dispatch system that shortens waiting time. • Prioritization of vulnerable families for faster rescues.	• Evacuation centers update capacity, resources, and safety status in real time.	• Post-disaster recovery system with updates on financial aid, relief goods, and rebuilding programs. • Guidance for families on safe rebuilding practices and future preparedness.

STORY BOARD:

Scenario: Typhoon "Mammamia" Approaches Barangay Maligaya

Barangay: Barangay Igualdad, Zone 5, Naga City, high-risk area.
Characters:
Randy's Family: **Ana** - Randy's Wife, **Erlinda** - Randy's Mother, **Ron** - Randy's Father, **Chin & Chad** - Randy's Kids.
Date/Time: 5:00 PM, Thursday. Typhoon "Mammamia" is expected to make landfall in 5 hours.



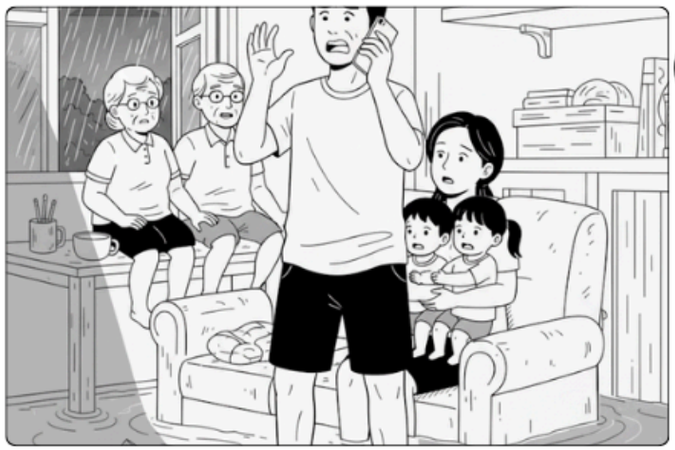
Randy came home after his day's work as a tricycle driver. As he neared his house, his neighbor informed him about an approaching typhoon.



Once inside, Randy didn't rest. He immediately started preparing by moving important belongings to higher shelves.



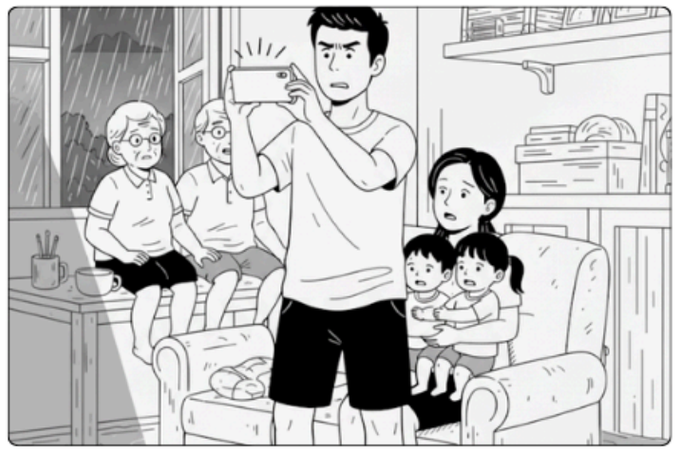
After a few minutes, he finished and finally sat down to rest with his wife. Soon, heavy rain began pouring outside.



Within just a few hours, floodwaters entered their home since their area is flood-prone. Randy prepared to call the hotline for evacuation help.



He tried calling the hotline several times, but no one answered. Desperate, he decided to ask for help through social media instead.



He took a video of his family situation and post it online pleading for help.



The water suddenly rose higher, forcing Randy to bring his family up to the roof. He used a tarpaulin to shield them from the rain and brought a small supply of food.



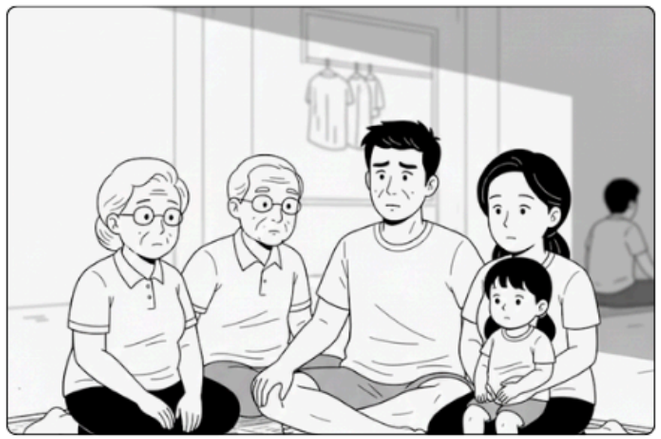
After hours of waiting, rescuers finally arrived. The family felt relief as they boarded a rescue boat.



On the way to the nearest evacuation center, travel was delayed because of blocked and flooded roads caused by damaged houses and debris.



They arrived at a crowded evacuation center filled with other evacuees. Randy registered their family at a table with barangay personnel.



They settled in a small space on the floor with a mat. They stayed there for the rest of the week.



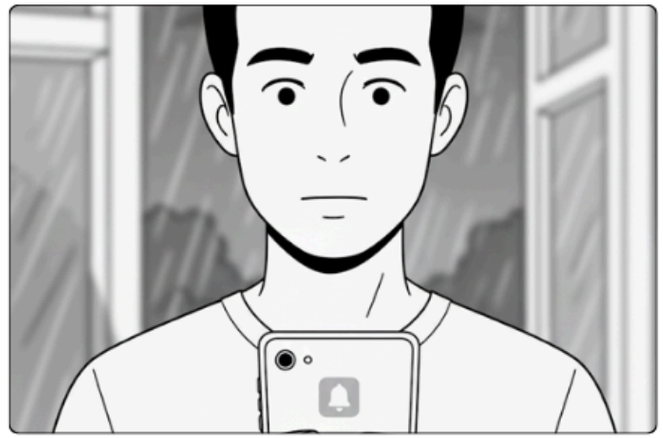
The family remained in the center for two weeks since floodwaters took long to subside. Eventually, they returned home only to find their house damaged.



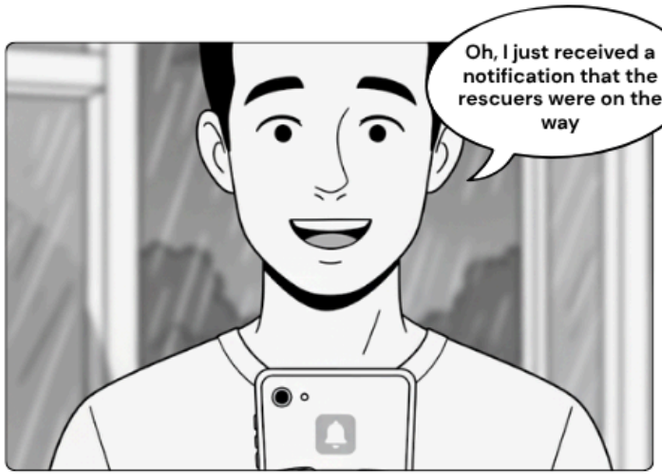
Randy was repairing the house while his wife, beside him, showed something on her phone.



Months later, another typhoon hit. Their area flooded again, forcing them to evacuate. This time, Randy was able to use the NagaRescue app



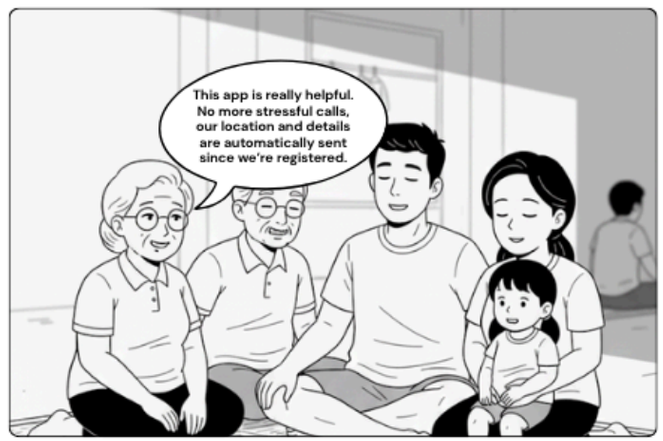
Within minutes, he received a notification that his request was acknowledged and they just had to wait.



After an hour, Randy received a message confirming rescuers were on the way. He told his wife the good news.



This time, the rescue was smooth. Thanks to guided safe routes from the app, rescuers had no trouble reaching them. At the evacuation center, Randy scanned a QR code on his phone during registration.



The family sat together on a mat all safe. They talked about how reliable the new app was.

USER & SYSTEM REQUIREMENTS SPECIFICATION

NAGARESCUE is a community-based disaster management system that connects the CDRRMO, barangays, rescuers, and residents to ensure faster and more organized emergency response. The following outlines the user, functional, and non-functional requirements for each user group.

USER 1: ComCen Staff (CDRRMO Naga Operators)

ComCen staff are the city-level disaster management operators responsible for monitoring barangay-submitted rescue requests, tracking ongoing rescue operations, and ensuring proper coordination between barangays and the CDRRMO. They act as overseers who step in only when a situation exceeds the barangay's capacity.

USER REQUIREMENTS

Must be able to view and monitor all barangay-submitted rescue requests, track rescuer deployment and operation status, and oversee city-wide disaster coordination.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
  - **FR1.1:** The system must receive and display barangay-submitted rescue requests and escalations in a centralized dashboard once data is pushed by the barangay.
  - **FR1.2:** The system must display complete rescue request details, including the resident's personal information, exact location, number of affected household members, and any identified vulnerabilities (e.g., elderly, children, or persons with disabilities).
  - **FR1.3:** The system must provide search and filter options by barangay.
  - **FR1.4:** The system must automatically update the ComCen dashboard when rescuers mark tasks as completed.

- **Non-Functional Requirements:**
  - **NF1.1:** The dashboard must update automatically without manual refresh.
  - **NF1.2:** The system must remain stable and responsive even when handling multiple simultaneous requests.
  - **NF1.3:** All tracking data and communications must be securely transmitted and encrypted.
  - **NF1.4:** Access must be restricted to authorized ComCen personnel only.

USER 2: Barangay Disaster Risk Reduction and Management Council (BDRRMC)

HCI TEMPLATE

The BDRRMC serves as the first responder to local emergencies, responsible for managing requests, monitoring barangay-level rescuers, and escalating cases beyond their capacity to the ComCen.

USER REQUIREMENTS

Must be able to manage, monitor, and escalate emergency requests within their barangay to ensure efficient local disaster response.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
  - **FR2.1:** The system must allow BDRRMC to view all active and completed emergency requests in their barangay.
  - **FR2.2:** The system must display complete rescue request details, including resident's personal information, exact location, household members, and identified vulnerabilities.
  - **FR2.3:** The system must enable filtering and sorting of requests based on location, time, or status.
  - **FR2.4:** The system must display assigned rescuers and their current status for each request.
  - **FR2.5:** The system must allow BDRRMC to escalate requests to the ComCen if they exceed barangay capacity.
  - **FR2.6:** The system must track escalation status and response time for efficiency monitoring.

- **Non-Functional Requirements:**
  - **NF2.1:** The dashboard must update automatically without manual refresh.
  - **NF2.2:** The system must remain stable and responsive under multiple simultaneous requests.
  - **NF2.3:** All data transfers, escalations, and communications must be secure and reliable.
  - **NF2.4:** The system must ensure data accuracy and confidentiality when displaying resident or request details.
  - **NF2.5:** Access must be restricted to authorized BDRRMC personnel only.
  - **NF2.6:** The system must provide a mobile-friendly interface or dedicated mobile app for accessibility during emergencies.

USER 3: Naga Residents (Requesters / Victims)

Naga residents are individuals directly affected by disasters who need immediate help. They use the mobile application to request assistance, monitor rescue updates, and ensure faster response during emergencies.

USER REQUIREMENTS

Must be able to pre-register, send emergency requests, and receive updates through a mobile application to ensure timely and efficient rescue response.

SYSTEM REQUIREMENTS

HCI TEMPLATE

- **Functional Requirements:**
  - **FR4.1:** The system must allow residents to pre-register their personal and household information for faster response during emergencies.
  - **FR4.2:** The system must allow one household representative to register and include multiple family members under a single household account.
  - **FR4.3:** The system must prevent duplicate registrations by ensuring each individual's name can only be registered once.
  - **FR4.4:** The system must allow residents to update their household or member information as needed.
  - **FR4.5:** The system must enable residents to send emergency requests with location and number of people affected.
  - **FR4.6:** The system must allow modification of the affected household list before submission without altering the main registration data.
  - **FR4.7:** The system must allow temporary inclusion of unregistered individuals (e.g., visitors or neighbors) in SOS requests.
  - **FR4.8:** The system must provide real-time updates on request status (received, dispatched, completed).
  - **FR4.9:** The system must display the rescuer's live location and route during an ongoing rescue.
  -
- **Non-Functional Requirements:**
  - **NF4.1:** The system must provide a mobile-friendly interface or dedicated mobile app for accessibility during emergencies.
  - **NF4.2:** The interface must be simple and intuitive even under stressful conditions.
  - **NF4.3:** The system must refresh data automatically without manual input.
  - **NF4.4:** Data accuracy and protection must be ensured for all registered households.
  - **NF4.5:** Notifications and updates must be low-latency.
  - **NF4.6:** The system must operate reliably even under limited mobile network connectivity.

HCI TEMPLATE

S.A.R.U. Deployment Management System (SDMS)

Bryann Joshua Francisco

USER PERSONA:



Maria Leonora

CDRRMO COMCEN WORKER

Resilient | Dedicated | Empathetic

Age: 45

Location: Barangay Igualdad, Zone 5, Naga City, Camarines Sur, Philippines

Biography

Maria Leonora has been serving in the City Disaster Risk Reduction and Management Office (CDRRMO) for 15 years as a communication center worker. With her degree in Public Administration, she has extensive experience in disaster preparedness, coordination, and community safety. Known by colleagues as highly organized, dependable, and calm under pressure, Maria is often at the center of crisis operations, making critical decisions when lives are at stake.

Her role requires her to oversee coordination between barangay officials, responders, and evacuation center officials during typhoons, floods, and other emergencies.

Challenges

- Calls can exceed the limit (2-3 minutes) and be lengthy
- Chance of being overwhelmed by high call volumes during disasters, leading to stress and slower coordination.
- Lack of a centralized system, resulting in fragmented communication across LGUs, barangays, and responders.

Technology Proficiency

- Proficient in using standard office tools (MS Office, spreadsheets, and reporting systems).
- Experienced with communication tools such as two-way radios, mobile phones, and email.
- Familiar with Messenger and group chats for coordination with barangay officials.

Goal & Needs

- Efficiently coordinate rescue operations with faster decision-making.
- Access a centralized system that shows all rescue requests, evacuation center status, and responder availability.
- Reduce delays in dispatching help during typhoons and floods.
- Promote proactiveness among citizens to lessen the disaster risk.
- Maintain clear and reliable communication with LGUs, barangays, and rescue teams even under high pressure.

Personality



JOURNEY MAP:



Maria Leonora

Age: 45
Occupation: CDRRMO COMCEN Worker
Location: Barangay Igualdad, Zone 5, Naga City

PERSONALITY

- Introverted: 4/10
- Cautious: 8/10
- Anxious: 5/10
- Community-Oriented: 8/10

GOALS & NEEDS

- Efficiently coordinate rescue operations with faster decision-making.
- Access a centralized system that shows all rescue requests, evacuation center status, and responder availability.
- Reduce delays in dispatching help during typhoons and floods.
- Promote proactiveness among citizens to lessen the disaster risk.
- Maintain clear and reliable communication with LGUs, barangays, and rescue teams even under high pressure.

CHALLENGES/ PAIN POINTS

- Calls can exceed the limit (2-3 minutes) and be lengthy
- Chance of being overwhelmed by high call volumes during disasters, leading to stress and slower coordination.
- Lack of a centralized system, resulting in fragmented communication across LGUs, barangays, and responders.

TECHNOLOGY PROFICIENCY

She is proficient in MS Office, spreadsheets, and reporting tools. She is also skilled in using communication tools like radios, phones, email, and Messenger for coordination.

Maria during before, during, and after Typhoon Kristine

PHASES	Pre-Disaster Monitoring	Early Impact (Heavy Rain Starts)	Peak of the Storm	Rescue Coordination Overload	Post-Storm Assessment
ACTIONS	Maria monitors weather updates from PAGASA and checks reports from barangays about their preparedness. She reviews contact lists, equipment availability, and standby teams.	As heavy rain begins, Maria starts receiving continuous calls from residents and barangay officials reporting rising flood levels. She takes notes manually while coordinating with response teams through calls and messages.	Floodwaters rise rapidly due to the lack of EWS and evacuation measures. Maria receives an overwhelming number of distress calls. With only four operators on duty, they struggle to accommodate all callers. Some lines stay busy for several minutes, and many calls go unanswered. She also finds it difficult to track which areas rescuers have already reached.	Maria juggles multiple channels — phone calls, Messenger chats, and barangay radio reports — to direct limited rescue teams. Many requests overlap, and verifying each one slows down response time.	When the storm subsides, Maria gathers reports from barangays about rescued families, damages, and supplies used. She spends hours compiling data manually for the city's disaster report.
EMOTIONS	She feels focused yet slightly anxious, knowing that the typhoon's strength could overwhelm their current system.	She feels overwhelmed and pressured, afraid of missing urgent rescue calls due to the high volume.	She feels pressured and frustrated, worried that people waiting for rescue might lose hope.	She feels overwhelmed but stays determined to coordinate as efficiently as possible despite limited manpower.	She feels relieved that major rescues are done but exhausted from the paperwork and manual tracking.
SOLUTION IDEAS	A centralized digital system that automatically displays weather alerts, barangay readiness levels, and the live status of standby teams to improve coordination even before the storm hits.	Create a system that has a real-time reporting dashboard that automatically logs and prioritizes rescue requests based on urgency and location.	Develop a system where victims can press an SOS button to automatically send a rescue request without calling. The system should log all requests, show their real-time status, and allow them to track and prioritize rescues based on severity and location.	Create a centralized system that displays all rescue requests and real-time updates from responders, allowing Maria to easily monitor ongoing operations and prioritize urgent cases without relying on hotline calls or social media alerts.	Automate post-disaster reporting and data collection through a unified system that generates summaries and maps automatically.

STORY BOARD:

Scenario: Typhoon "Mammamia" Approaches Noga City

City/Municipality: Noga City, a disaster prone area in Region V
Key Personnel:
CDRRMO NAGA COMCEN WORKER: Maria Leonora, her role requires her to oversee coordination between barangay officials, responders, and evacuation center officials during typhoons, floods, and other emergencies.
Date/Time: 12:00 PM, Thursday. Typhoon "Mammamia" is expected to make landfall within 24 hours. The alert is CPA Charlie (Mandatory Evacuation).



Maria is at stand-by in the CDRRMO office. She is monitoring PAGASA updates and are waiting for news on flood-prone barangays.



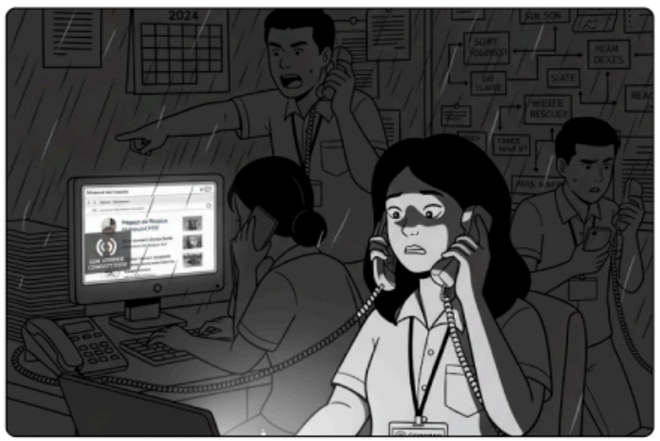
Due to the lack of an EWS, unprepared citizens begin to ask for rescues.



With the flood continuously drowning barangays, Maria and her co-workers continue to struggle, some requests are even unanswered due to the influx of volume.



Meanwhile, rescuers often get lost and fail to contact both Maria and the victims for updates due to the flood and other environmental hindrances



Maria is left in the dark about the status of some requests, yet she still continues to search on social media platforms for posts about stranded victims even though there is slow internet connection



As the storm subsides, Maria needs to do a lot of manual paperwork to check on evacuation centers, request details and status, and look for trapped individuals and



Maria enters the NAGARESCUE app and monitors the disaster-prone areas using the digital map based on past data.



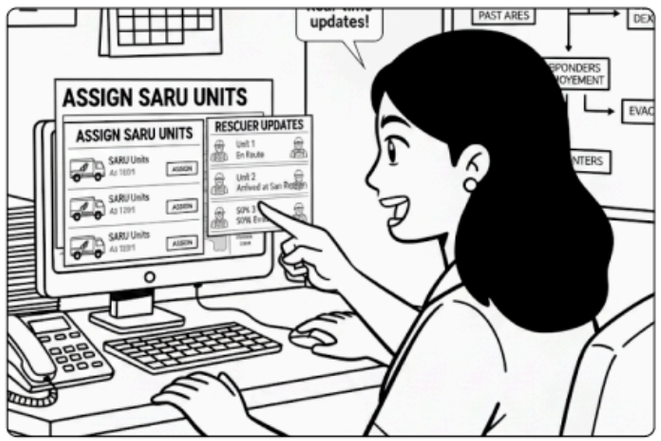
Through the app she recieves updates from barangay point persons about the status of zones in baranggays.



She urges the barnagays to utilize the early warning system through the NAGARESCUE application to warn the citizens on disaster-prone areas.



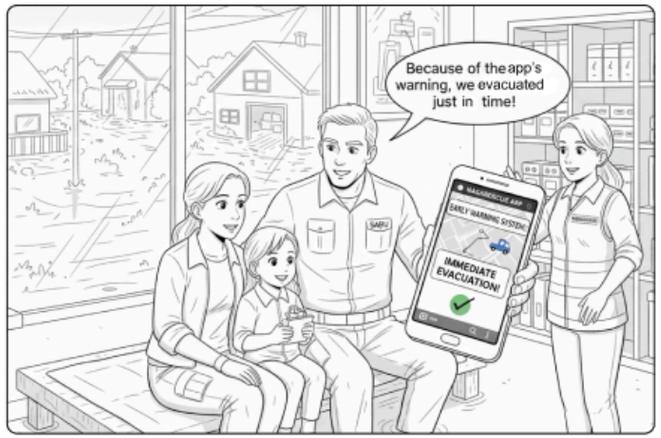
Through the application she recieves rescue requests. This time the rescue contains the details of the victims instead of having to call them for 2-3 minutes...



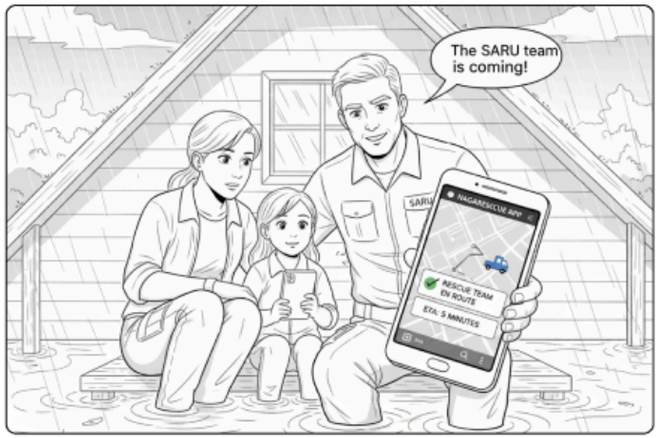
Through the app, she assigns SARU units to the requests, even managing to see updates from the rescuers themselves.



The rescuers see the most convenient route towards the destination, the app even warns them of what routes are unsafe.



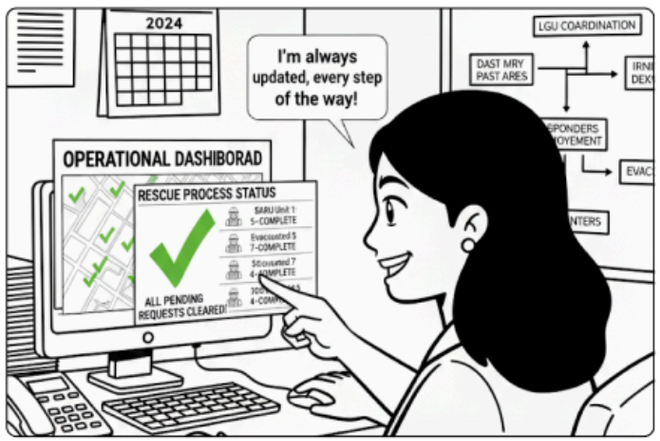
Citizens are saved in advance due to the EWS in the app.



The citizens are given updates about the status of their request.



The rescuers manage to save victims more efficiently because of the app



Maria is always updated about the rescue status, making her feel more at ease and helping her in documentation.



Maria and the other three ComCen workers feel more at ease because they can process each request proeperly and can return to requests that they overlooked



Documenting disaster details are done easier because of the application...

USER & SYSTEM REQUIREMENTS SPECIFICATION

USER 1: ComCen Staff (CDRRMO)

The City Disaster Risk Reduction Management Office (CDRRMO) is the LGU that oversees the practice of DRR in the city of Naga. As part of their practice, they provide rescue services to those who are in need. An important component in this service is the communication center (ComCen). The ComCen is composed of 3 people every 24 hours; they take the calls of the citizens, get their details and deploy the SARU units to the citizen's location.

USER REQUIREMENTS

The ComCen staff must be able to receive a summarized detail of the citizen: name, age, location, sign of injuries/sickness, status, family members, and location. These requests must be stored and presented in a neat UI so that the staff can view all requests and consider them. The ComCen must be able to see the list of SARU units and be able to deploy them on the requests that they received and see the details of the rescue process.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
  - **FR1.1:** The system must display rescue requests with summarized details.
  - **FR1.2:** The system must display the list of SARU units and their equipment (personnel, PPE, life vests, radios, headlights, etc.).
  - **FR1.3:** The system must allow CRUD operations on SARU unit records.
    FR1.4: The system must allow ComCen staff to assign and deploy SARU units to rescue requests.
  - **FR1.5:** The system must display request locations on a digital map using color-coded indicators considers the children, pregnant, unconscious, elderly, disabled, and wounded.
  - **FR1.6:** The system must allow ComCen staff to mark flooded areas on the map.
  - **FR1.7:** The system must visually display flooded or blocked areas based on reports.
  - **FR1.8:** The system must record and organize citizen details for each rescue.
  - **FR1.9:** The system must receive notification updates from SARU units.
    FR1.10: The system must track SARU units and their current status (available, en route, completed).
  - **FR1.11:** The system must show the current location of responders; if offline, it must display at least the rescue site.
  - **FR1.12:** The system must support searching and filtering rescue requests by urgency, barangay..
  - **FR1.13:** The system must support searching and filtering SARU units by status (active, standby).
  - **FR1.14:** The system must receive registration details of responders.
  - **FR1.15:** The system must allow ComCen to confirm responder registrations.
  - **FR1.16:** The system must allow ComCen members to register based on valid credentials (verified by an admin).
  - **FR1.17:** The system must allow ComCen members to log in using username and password.

TEMPLATE

- **Non-Functional Requirements:**
    - **NF1.1:** The system's UI must be clean, organized, and intuitive.
    - **NF1.2:** The system must perform efficiently in low-latency environments.
    - **NF1.3:** The system must encrypt citizen data securely.

USER 2: SARU Responder (CDRRMO)

The CDRRMO is the LGU that oversees the practice of DRR in the city of Naga. As part of their practice, they provide rescue services to those who are in need. An important component in this service is the SARU unit. The SARU unit works under the Operations and Warnings Department of the CDRRMO. It is their responsibility to go to the location of the caller based on the details that the ComCen gave them and conduct the rescue operation as well as submit reports back to the ComCen that contain the details of the operation.

USER REQUIREMENTS

The system must help them in knowing the details of the request based on the updated and accurate information shared by the citizen through the CDRRMO. The SARU responder must be able to use the system's routing feature in order to travel to the citizen's location faster. The system should also allow the responder to send notification updates to the ComCen and the citizen.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
    - **FR2.1:** The system must allow SARU responders to register accounts for verification.
    - **FR2.2:** The system must allow secure login for SARU responders.
      FR2.3: The system must display assigned operations from ComCen.
    - **FR2.4:** The system must allow responders to update their operational status (available, en route, on-site, rescued, completed).
      FR2.5: The system must enable responders to send notification updates to ComCen and citizens.
    - **FR2.6:** The system must automatically timestamp key actions (dispatch, arrival, completion).
    - **FR2.7:** The system must allow responders to mark their post-mission availability (active, standby, rest).
    - **FR2.8:** The system must provide a map-based routing feature for fastest and safest travel routes.
    - **FR2.9:** The system must display current road conditions based on ComCen updates.

    - **FR2.10:** The system must allow responders to submit post-operation reports (rescued individuals, injuries, materials used).

- **Non-Functional Requirements**
    - **NF2.1:** The system must be optimized for mobile devices (smartphones, tablets).

HQ TEMPLATE

- **NF2.2:** The system must work effectively offline or with unstable connectivity.
- **NF2.3:** The mobile UI must be minimal, responsive, and easy to navigate under stress.
- **NF2.4:** The system must provide visual and audible alerts for new or urgent messages.

USER 3: BDRRMC

The Barangay Disaster Risk Reduction Management Council is the branch of the CDRRMO in barangays. Their task is to respond to the requests and forward the requests to the CDRRMO.

USER REQUIREMENTS

The system must be able to receive localized requests and allow the BDRRMC to forward any request that they cannot cater to or attend to the CDRRMO so that these requests can be answered.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
  - **FR3.1:** The system must allow BDRRMC staff to securely log in to their barangay account.
  - **FR3.2:** The system must allow BDRRMC staff to register their credentials for account creation.
  - **FR3.3:** The system must allow a pre-existing BDRRMC account to verify new registrations.
  - **FR3.4:** The system must allow recording and viewing of localized rescue requests.
  - **FR3.5:** The system must display requests on a color-coded digital map (children, pregnant, elderly, disabled, etc.).
  - **FR3.6:** The system must allow BDRRMC to respond to or escalate requests to the CDRRMO.
  - **FR3.7:** The system must include a simple routing feature for efficient navigation.
  - **FR3.8:** The system must show all ongoing and completed barangay rescue operations in a dashboard.
  - **FR3.9:** The system must receive updates and notifications from CDRRMO regarding escalated cases.
  - **FR3.10:** The system must allow BDRRMC to generate summary reports of incidents and responses.
  - **FR3.11:** The system must store request and operation histories for review and accountability.

- **Non-Functional Requirements**
  - **NF3.1:** The system must be optimized for mobile use in the field.
  - **NF3.2:** The system must function offline or with limited connectivity.
  - **NF3.3:** The mobile interface must be minimal, responsive, and user-friendly.
  - **NF3.4:** The system must provide visual and audible alerts for new or urgent messages.

TEMPLATE

Evacuation Monitoring System (EMS)

Gabriel David Diego

USER PERSONA:



Pedro Penduco

Evacuation Personnel

Responsible | Social | Focused

Age: 40

Location: Zone 1, Mabolo Naga City, Camarines Sur, Bicol, Philippines

Biography

Pedro Penduco is a 40-year-old barangay worker in Naga City who runs the local evacuation center. When storms come, he handles all the logistics: preparing food, water, blankets, keeping basic inventory, registering evacuees by hand. He's committed and works hard, but the centers often become overcrowded and supplies run out fast, Kristine made that very clear.

He's trying to protect his community under really tough conditions. Even with limited resources, unreliable systems (power/internet), and inadequate shelter facilities, Pedro always aims to do better: improve our preparedness, coordinate response, and make sure people are safe, especially those who need extra care.

Challenges

- The center lacks a dependable digital system, and power or internet often fail.
- Staff and volunteers are few and stretched thin during disasters, and the shelters (often schools or gyms) have limited space and poor facilities.
- Because of these challenges, he rarely has accurate data on who's in the center, who needs special care, or whether the available resources can serve everyone.

Technology Proficiency

- Learns best when things are simple, clear, and hands-on.
- Has a lot of experience with smartphones: can use messaging and streaming apps.
- Gets stuck when power or internet don't work.
- Basic experience with computer; having trouble using spreadsheets and dashboards.

Goal & Needs

- Ensure evacuee safety, health, and special-needs care by having low-latency tracking of people and supplies
- Maintain adequate resources and minimize waste through alerts when stock or space is low
- Prevent overcrowding and optimize accommodation by enforcing clear rules for admission and overflow
- Coordinate with rescuers, local government, and relief agencies using reliable communication, reporting, and a safe shelter lavout with zones for medical. isolation.

Personality



JOURNEY MAP:



Pedro Penduco

AGE: 40
GENDER: MALE
LOCATION: Zone 1, Mabolo Naga City

PERSONALITY

- Introverted: 8/10
- Analytical: 9/10
- Loyal: 9/10
- Patient: 8/10

GOALS

- Ensure she knows the total product sales at the end of her shift.
- Ensure that the cash in the money box matches the recorded sales to avoid discrepancies.
- Maintain an accurate count of remaining stock

NEEDS

He needs fast tracking of people and supplies, alerts for shortages, and clear rules for admissions. He also needs dependable communication, concise reporting, and shelter spaces arranged safely (medical, isolation, women/children).

CHALLENGES/ PAIN POINTS

- The center lacks a dependable digital system, and power or internet often fail.
- Staff and volunteers are few and stretched thin during disasters, and the shelters (often schools or gyms) have limited space and poor facilities.
- Because of these challenges, he rarely has accurate data on who's in the center, who needs special care, or whether the available resources can serve everyone.

Managing Evacuation Center Journey Map

PHASES	Preparation / Pre-Response	Evacuation Decision	Shelter & Assistance	Recovery / Return
ACTIONS	• Monitor hazard / weather alerts • Identify suitable shelters in advance • Preposition supplies, plan layout • Engage volunteers & coordinate with LGU and rescuers	• Activate evacuation plan • Issue warnings and messages to residents • Direct people to move safely to the center • Manage transportation & arrival routing	He registers evacuees and gathers their data, assigns them to zones or beds, distributes supplies, monitors occupancy and resource usage, identifies and prioritizes vulnerable people, coordinates with external agencies for support, and adjusts overflow or opens extra spaces as needed.	• Evacuees depart or return to home • Close or scale down the center • Tally resource usage • Debrief with team, identify lessons learned • Report to LGU / stakeholders
EMOTIONS	Uneasiness 😟	Tension(Risk of Chaos need for clarity in communication) 😟	Overwhelmed because of the number of tasks	Relief (Typhoon is gawn) 😎
SOLUTION IDEAS	Integration of Predefined shelter maps(official and community-based) & resource staging Volunteer / role assignment tool Stock prepositioning system	Simple warning / messaging system Routing maps, traffic control planning Coordination with rescue teams, transport services	Real-time occupancy / inventory dashboard Alerting for low resource / nearing capacity Admission / overflow protocols built into the system Vulnerability tagging & sorting Communication / coordination interface with rescuers / LGU Layout planner with safe spacing & zones	Post-event reports & analytics Feedback collection from evacuees & staff Revision of procedures & preparedness plan Facility improvement planning

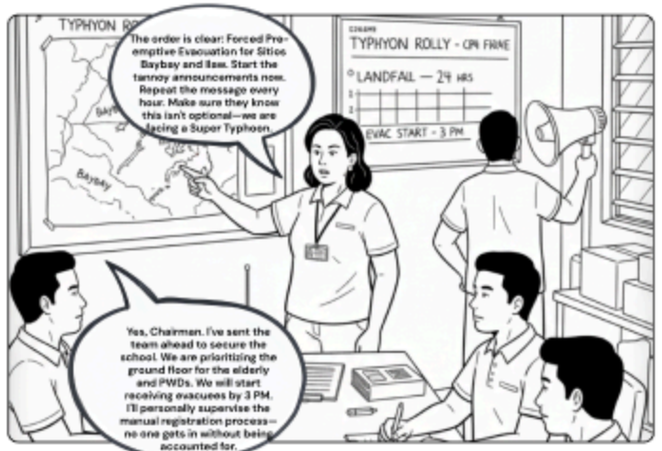
STORY BOARD:

Scenario: Typhoon "Mammamia" Approaches Barangay Maligaya

Barangay: Maligaya, a coastal, high-risk area.
Key Personnel:
Punong Barangay: Aling Minda, head of the BDRRMC.
Camp Manager: Sir Ben, head of the Evacuation Management Team.
Date/Time: 12:00 PM, Thursday. Typhoon "Mammamia" is expected to make landfall in 24 hours. The alert is CPA Charlie (Mandatory Evacuation).



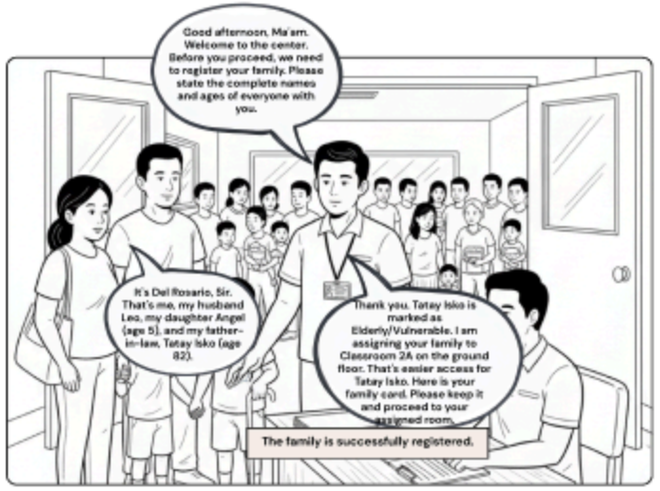
Chairwoman is concluding the emergency meeting with her cluster heads.



Giving orders for a forced preemptive evacuation for Super Typhoon Mammamia, prioritizing the elderly and PWDs, and ensuring everyone is accounted for.



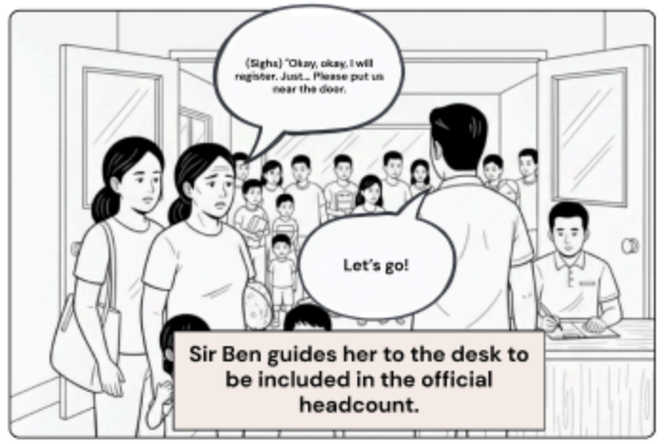
Chairwoman orders forced typhoon evacuation, assigning Sir Ben to lead.



A family with an elderly member registers at the evacuation center.



An official explains why registration is mandatory for their safety.



She relents and registers, asking to be near the door.



The Camp worker introduces an app to streamline future evacuation registrations.



The app can set evacuation center capacity to manage supplies.



The Camp Manager decides to use the app to avoid fut overcrowding.



The chairman arranged a meeting for the residents and camp manager of their community to introduce the application that they found.



The residents are confused why the chairman arranged a meeting to discuss the app that the Camp Manager found.



The residents realized that the application will be a lot of help to them.



After a few months, another typhoon landed on their community.



This time, they are not manually counting but using an ID from the NAGARescue with the list of their family.



The solo resident used his family member ID instead of family ID because he is not with his family.



The camp manager is extremely satisfied with the capabilities of the application.



The camp worker who found the app is also very proud about the app that he found.



The camp manager decided to stick with the application for life. **DU DUUUNN**

USER & SYSTEM REQUIREMENTS SPECIFICATION

USER 1: BDRRMC (Camp Manager)
The camp manager under the BDRRMC is the primary authority responsible for the overall management and operation of a designated evacuation site. Their main goal is to ensure the safety, security, and well-being of all evacuees.

USER REQUIREMENTS
The system must provide a unified platform for managing all core evacuation center operations, allowing the user to register and track the evacuee population, and report incidents to the ComCen.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
    - **FR1.1: Head Counting**: The system must be able to validate personal information using the FamilyID(this is from another module).
    - **FR1.2: Evacuee Status Management**: The system must provide functions to manage evacuee records, including updating their status (e.g., checked-in).
    - **FR 1.3: Population Dashboard:** The system must automatically calculate and display a real-time total of the evacuee population and show this count relative to the center's maximum capacity (e.g., "750 / 1000").
    - **FR1.4: Report Exporting:** The system must be able to export required reports inside the evacuation center(e.g., number of families, individuals).

- **Non-Functional Requirements:**
    - **NF1.1: Usability**: The interface is simple, intuitive, and optimized for rapid data entry under high-stress conditions.
    - **NF1.2: Data Integrity & Synchronization**: All locally stored data automatically syncs with the central ComCen server when connectivity is established, with all transactions timestamped for accountability.
    - **NF1.3: Security:** All personally identifiable evacuee data is encrypted both locally and during transmission.
    - **NF1.4: Performance:** Data transmission is optimized to function effectively even on low-bandwidth or unstable network conditions.

USER 2: Naga Citizen / Victims
An individual or family head affected by a disaster who is seeking shelter, basic necessities, and critical information. Their main goal is to find safety, register for accountability, receive aid, and stay informed about the situation.

USER REQUIREMENTS:

HCI TEMPLATE

The system must provide a simple and accessible mobile platform for evacuees to locate designated evacuation centers, register themselves and their family, receive official announcements, and submit personal assistance requests.

SYSTEM REQUIREMENTS

- **Functional Requirements:**
    - **FR2.1: Evacuation Center** Locator: The system must display a list or map of nearby, open evacuation centers.
    - **FR2.2: Center Status Display:** The system must display the current population status (e.g., "Open," "At Capacity") for selected evacuation centers.
    - **FR2.3: Self-Registration:** The system must allow an evacuee (as a family head) to submit their personal and family members' details to pre-register or check-in at a center.
    - **FR2.4: Offline Information Access:** The system must cache critical information (e.g., safety guidelines, first-aid info, evacuation center addresses) for offline viewing.

- **Non-Functional Requirements:**
    - **NF2.1: Usability & Accessibility:** The interface must be extremely simple, intuitive, and easy to use for people who are not tech-savvy, under high stress, and in low-light conditions (e.g., clear icons, large fonts). It should also be accessible to people with disabilities (WCAG compliance).
    - **NF2.2: Reliability (Offline Caching):** The system must cache critical information (like maps, and safety guides) so it can be accessed even if the user loses their data connection.
    - **NF2.3: Data Integrity & Synchronization:** User-submitted data (like registrations or requests) must be queued and sent automatically once a minimal network connection is detected.
    - **NF2.4: Security & Privacy:** The system must encrypt and protect the user's personal data (PII) to prevent unauthorized access and ensure their privacy and dignity.
    - **NF2.5: Performance**: The application must be lightweight, consume minimal data, and load quickly, making it functional on low-end mobile devices and very poor network conditions (e.g., 2G/3G).

USER 3: CDRRMO (ComCen) Operator

An Organization who oversees the city and the one tasked with all the decision making.

USER REQUIREMENTS:

The system must provide a centralized, real-time "Common Operational Picture" (COP) that aggregates data from all evacuation centers and community shelters. This will allow the user to monitor city-wide evacuee populations, track resource levels across all sites, and make strategic decisions about resource dispatch and evacuee redistribution.

SYSTEM REQUIREMENTS:

HCI TEMPLATE

- **Functional Requirements:**
  - **FR3.1: City-Wide Dashboard:** The system must display a map and a consolidated list of all evacuation centers and registered community shelters, color-coded by their current capacity (e.g., Green: 0-79%, Red: 80-100%, Black: Over Capacity).
  - **FR3.2: Aggregate Data View**: The system must display city-wide totals: Total Evacuees, and Total Available Capacity.
  - **FR3.3: Evacuee Redistribution Tool:** The system must allow the user to identify nearby, under-utilized shelters to redirect evacuees from an overcrowded center.
  - **FR3.4: Consolidated Report Generation**: The system must generate and export consolidated situational reports in DSWD/NDRRMC-compatible formats, as specified in the proposal.

- **Non-Functional Requirements:**
  - **NF3.1:** High Availability: The central dashboard must be web-based and have low-latency uptime during a disaster.
  - **NF3.2:** Data Visualization: Data must be presented visually (maps, graphs, charts) for rapid, at-a-glance decision-making.
  - **NF3.3:** Concurrency: The system must be able to handle simultaneous data syncs from hundreds of Camp Managers.

HCI TEMPLATE